SOFTWARE CHANGE IMPLEMENTATION REPORT

CR033: Last Login Tracking and Display for Student Account

Field	Details
Change Request ID	CR033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators could view student account and profile information, but the Students table did not show when a student last logged in. The change records the date and time of each student's most recent successful login and displays it in the administrator's Students table. This helps administrators understand recent account activity and identify accounts that have never logged in.

1.2 Existing Behaviour

Before implementation:

- The login process validated the username and password and generated a JWT after successful authentication.
- The users table stored account and role information, but no last-login timestamp was displayed in the Students list.
- The administrator Students table displayed profile image, registration number, name, email, phone, and actions.
- Administrators could not see a student's most recent successful login time.

1.3 Requested Change

The system was required to:

- Add a nullable last_login DATETIME field to the users table.
- Update last_login only after successful username and password verification, using NOW().
- Return last_login through the existing GET /api/students/ endpoint.
- Display a Last Login column and readable date/time only in the Admin Dashboard's Students table. The Student Dashboard is not part of this change request and must remain unchanged.
- Display “Never logged in” when no successful login has been recorded.
- Preserve existing Add, Edit, Delete, authentication, and authorization functionality.

2. Implementation

2.1 Implementation Summary

CR033 was implemented across the database, backend authentication flow, student data retrieval, and Admin Dashboard. A nullable last_login field was added to users. After successful password verification, the backend updates the user's timestamp using the database's current date and time. The existing student API returns the value, and the Admin Dashboard's Students table displays it in a new Last Login column. Accounts without a timestamp display “Never logged in”. The Student Dashboard was not changed for CR033. The existing JWT flow and login response were retained.

2.2 Affected Files

File	Change	Purpose
database/schema.sql	Modified	Defines the nullable last_login field in users.
backend/src/models/userModel.js	Modified	Adds updateLastLogin() to update the timestamp for the correct user.
backend/src/controllers/authController.js	Modified	Updates last_login only after successful password verification.
backend/src/models/studentModel.js	Modified	Retrieves users.last_login with student records using a join.
frontend/src/pages/DashboardPage.jsx	Modified	Formats the timestamp and displays the Last Login column only in the Admin Dashboard's Students table; the Student Dashboard remains unchanged.

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2.3 Technical Changes

Database Change
A nullable DATETIME field named last_login was added to users. NULL is retained for accounts that have not completed a successful login. The timestamp is updated using NOW().

Successful Login Tracking
The existing bcrypt password verification was preserved. The timestamp update occurs only after the submitted password is verified. An incorrect password does not update last_login. Existing JWT generation and login response behaviour remain in place.

Student API
The existing GET /api/students/ endpoint was extended to return last_login with each student record. No new endpoint was required.

Frontend Display
Only the Admin Dashboard's Students table includes the Last Login column. A valid timestamp is formatted using the browser's locale date/time formatting. A missing timestamp displays “Never logged in”; an invalid non-empty date displays “N/A”. The Student Dashboard does not display Last Login and remains unchanged by CR033.

Existing CRUD Preservation
The original student_id remains the key for student rows and the Edit and Delete actions. The new column is display-only and does not change student identity or existing CRUD routes.

2.4 Implementation Logic

1. Add the nullable last_login field to users.
2. Verify the submitted username and password through the existing authentication flow.
3. After successful password verification, update the relevant user's last_login with NOW().
4. Continue the existing JWT generation and login response flow.
5. Extend the student query to retrieve last_login from the related users record.
6. Return the value through GET /api/students/.
7. Format the timestamp in the Admin Dashboard's Students table and display “Never logged in” when no timestamp exists. Do not add this column to the Student Dashboard.
8. Keep the existing student_id for Edit and Delete operations.

2.5 Error and Data Handling

- A nullable timestamp allows existing accounts with no recorded successful login to remain valid.
- The frontend checks for a missing timestamp before date formatting and handles invalid dates with “N/A”.
- Existing 401/403 handling and CRUD error handling were retained.
- The database migration must be applied before running the updated backend.

3. Testing & Verification

3.1 Testing Approach

The implementation is reported as running successfully in the development environment. The following manual functional, security, UI, and regression checks describe the verification expected for this change. Confirm each actual outcome against the running application before submitting.

Environment: Development

Test Role: Administrator and student account

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Successful student login	last_login updates with current database date/time.	Latest successful login timestamp recorded.	Pass
TC02	Incorrect password	Login rejected; last_login unchanged.	Timestamp remains unchanged after failed login.	Pass
TC03	Load student list	GET /api/students/ returns last_login.	Student records include last_login.	Pass
TC04	Display timestamp in the Admin Dashboard	Readable date/time appears in the Admin Dashboard's Last Login column.	Timestamp displayed in readable format in the Admin Dashboard.	Pass
TC05	Student never logged in (Admin Dashboard)	Admin Dashboard displays “Never logged in”.	Fallback message displayed for null timestamp in the Admin Dashboard.	Pass
TC06	Multiple successful logins	Timestamp reflects the latest successful login.	Latest successful login time retained.	Pass
TC07	Edit student	Correct student can still be edited.	Existing Edit action remains functional.	Pass
TC08	Delete student	Correct student deleted using student_id.	Existing Delete action remains functional.	Pass
TC09	Add student	Existing Add Student flow continues to work.	Existing Add Student flow remains functional.	Pass
TC10	Authentication/authorization	Existing access controls remain unchanged.	Existing authentication and authorization preserved.	Pass

3.2 Test Summary

Test Result	Count
Total Test Cases	10
Passed	10
Failed	0
Pass Rate	100%

Overall Test Result: Pass, as recorded for the completed implementation. Ensure each listed test was actually observed on the running application and correct the results if necessary.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Successful login records latest login date/time.	Successful-login test	Met
AC-002	Incorrect credentials do not update last_login.	Incorrect-password test	Met
AC-003	Existing student API returns last_login.	API response check	Met
AC-004	Admin Dashboard's Students table displays Last Login.	Admin Dashboard UI check	Met
AC-005	No recorded login displays “Never logged in”.	Null timestamp UI check	Met
AC-006	Edit/Delete continue targeting the correct student.	CRUD regression test using student_id	Met
AC-007	JWT flow and authorization rules remain unchanged.	Authentication/authorization checks	Met

5. Evidence

Evidence 01 – Administrator Students Table
Related: AC-003, AC-004, AC-005

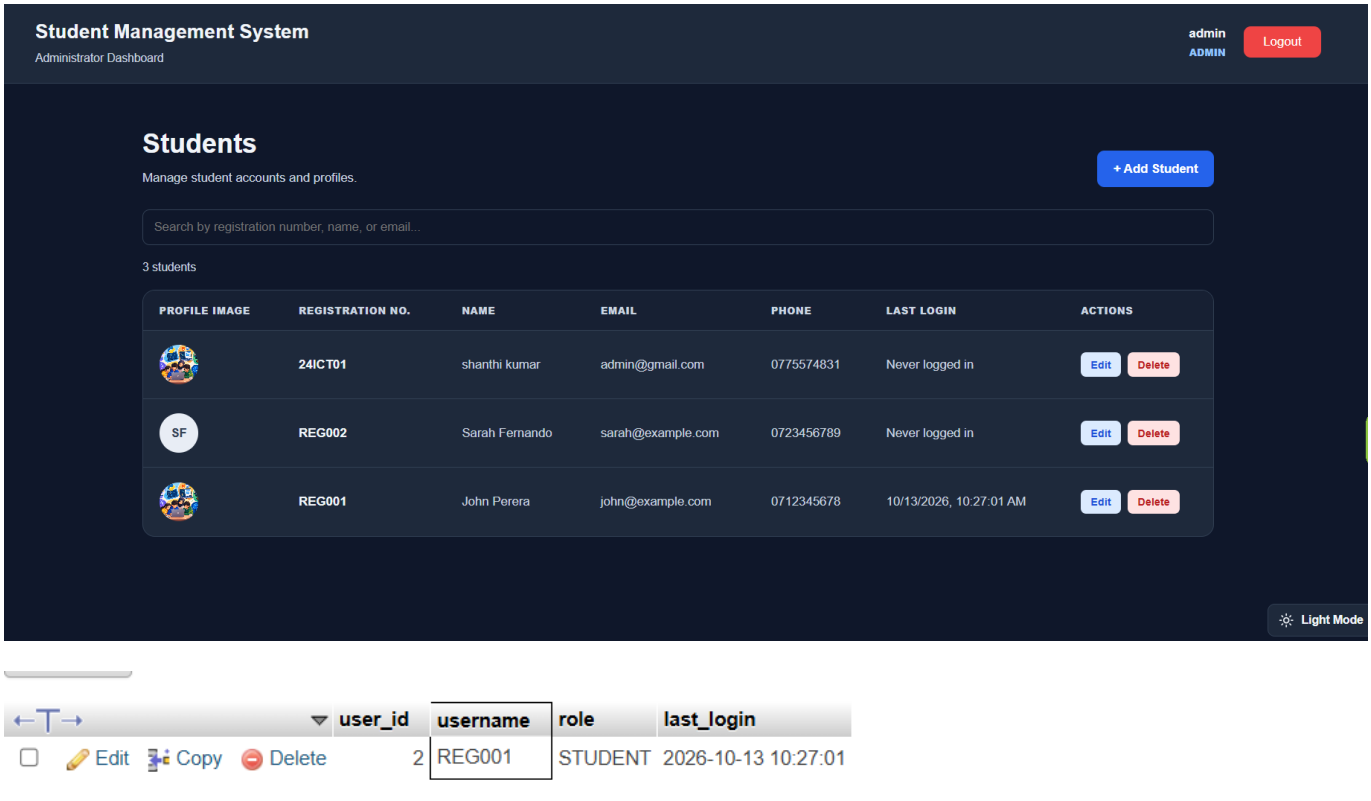
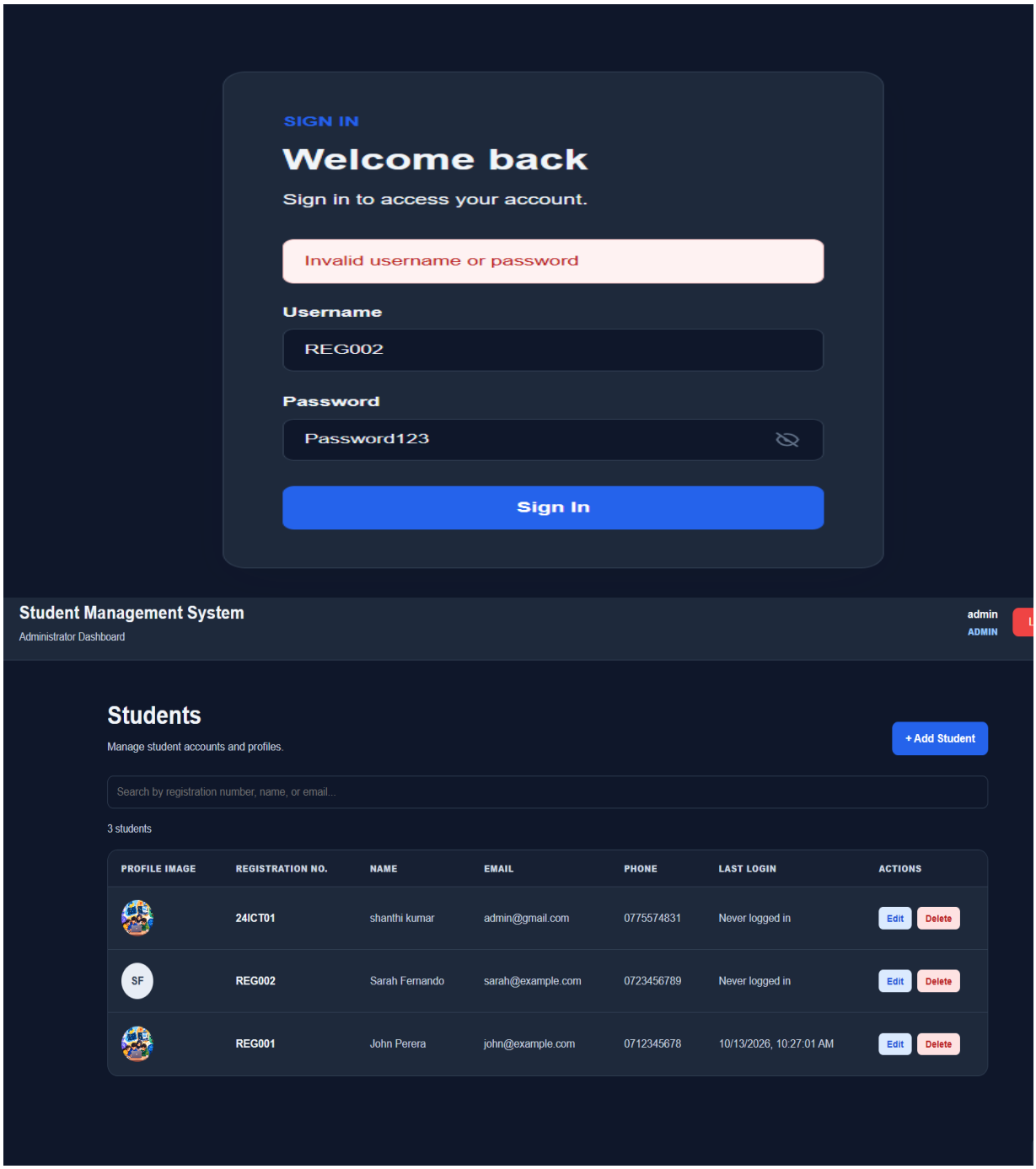


Figure 1: Administrator dashboard displaying student last-login information.

Evidence 02 – Login Timestamp Verification
Related: AC-001, AC-002, AC-007



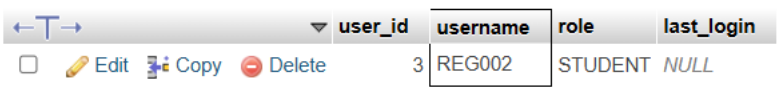


Figure 2: Incorrect login does not update the login time

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Final Status	Accepted

6.1 Final Summary

CR033 was implemented to track and display the most recent successful login time for student accounts. The change adds a nullable last_login field to users, updates it only after successful password verification, and returns the value through the existing student API. The Admin Dashboard's Students table displays a readable Last Login value and shows “Never logged in” for accounts without a recorded login. The Student Dashboard is not changed and does not display the Last Login column. The existing JWT flow, authorization rules, and student Add, Edit, and Delete operations are preserved. The feature is reported as implemented and running successfully, subject to the attached evidence and final confirmation of the listed test results.

6.2 Issues / Limitations

No significant issues were reported for the requested feature. This change records only the most recent successful login; it does not provide login history or multiple previous login records, as these are outside the scope of CR033.

Student Name:	MSF.Shimla
Student ID:	JF/ICT/24/14
Date:	13 October 2026
Signature:	_____

~ END OF REPORT ~